XAUUSD Bot: Internal Workflows and Trade Logic

This document outlines the step-by-step internal workflow of the XAUUSD Python Bot, from data fetching to final trade execution.

--- STAGE 1: DATA AGGREGATION & FILTERING ---

Every 60 seconds, the main ICT engine (bot_free.py -> ict_loop) wakes up.

1. The bot attempts to connect to MetaTrader 5 over Port 9999 to fetch the last 200 bars of pure broker OHLCV tick data.
2. If MT5 is disconnected, it forcefully pulls data from the Yahoo Finance API (GC=F).
3. The ForexFactory XML Guardian natively checks the time. If the current clock is within 30 minutes (before or after) of a Red Folder high-impact US economic event, the bot immediately suspends itself.
4. The Time Guardian verifies it is within the active ICT 'Kill Zone' hours.

--- STAGE 2: DETERMINISTIC LOGIC (SMART MONEY CONCEPTS) ---

Once the data is verified and macroeconomic events are cleared, the technical engine fires:

1. DAILY BIAS: Calculates the 50-period and 200-period Exponential Moving Averages (EMA) on the 1-Hour chart. If EMA50 > EMA200, the bias is strictly BULL.
2. ADX MOMENTUM: Calculates the Average Directional Index (ADX). If ADX is below 22, the market is ranging, and the bot suspends.
3. STRUCTURE: The bot scans the last 30 bars for a Fair Value Gap (FVG) or an algorithmic Liquidity Sweep exclusively in the direction of the daily bias.
4. QUALITY SCORE: If a setup is found, it calculates a Quality Score based on distance from Moving Averages, RSI divergence, and time of day. The setup must score over 40.

--- STAGE 3: THE AI INFERENCE GUARDIAN (XGBoost/Scikit-Learn) ---

If the deterministic engine generates a valid trade signal, it hands off control to the Machine Learning Guardian before executing the trade.

1. The AI module natively calculates 11 live mathematical features for the current cande (including distance from EMAs, live RSI, and body percentages).
2. It simultaneously runs an emergency background fetch of the US Dollar Index (DXY) and 10-Year Treasury Yields (^TNX), calculating their absolute 1-Hour percentage spike.
3. It packages these 11 data points and injects them into the pre-trained 'ml_gold_long' or 'ml_gold_short' .joblib model.
4. The AI returns a structural probability score from 0.0 to 1.0. If the score is under 0.65 (65%), the AI kills the trade.

--- STAGE 4: EXECUTION PROTOCOL (MetaTrader 5 Bridge) ---

If the AI approves the trade, execution protocols are engaged:

1. RISK MANAGEMENT: Stop Loss is dynamically calculated using 0.6 * ATR (Average True Range) to adjust for live market volatility.
2. TARGETS: Take Profit 1 is hardcoded mathematically to a hyper-aggressive 0.5R. Take Profit 2 is

extended to 3.5R.

3. POSITION SIZING: The Python bot constructs a raw command string (e.g., "LONG|1.20|0.60|4.20").

4. TCP SEND: It fires the string over TCP Socket to the EA attached inside MetaTrader 5.

5. THE MT5 EA: The MT5 EA catches the string, executes a Market Order, computes the exact Lot Size risking 1% of the account, and automatically sets the Break-Even trail once TP1 is hit.